A Novel Feature Vector Using Complex HRRP for Radar Target Recognition*

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Abstract. Radar high-resolution range profile (HRRP) has received intensive attention from the radar automatic recognition (RATR) community. Since the initial phase of a complex HRRP is strongly sensitive to target position variation, which is referred to as the initial phase sensitivity, only the amplitude information in the complex HRRP, what is called the real HRRP, is used for RATR. This paper proposes a novel feature extraction method for the complex HRRP. The extracted complex feature vector contains the difference phase information between range cells but no initial phase information in the complex HRRP. The recognition algorithms, frame-template-database establishment methods and preprocessing methods used in the real HRRP-based RATR can also be applied to the proposed complex feature vector-based RATR. The recognition experiments based on measured data show that the proposed complex feature vector can obtain better recognition performance than the real HRRP if only the cell interval parameters are proper.

1 Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS). It contains target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1~5]. According to the definition of HRRP given in the above literatures, an HRRP is a real vector, which is the amplitude of complex returned echo vector in baseband. Obviously, the phase information in the complex returned echo is not applied to recognition.

Similar to the scatterer distribution information contained in the real HRRP, the phase information contained in the complex HRRP should also be valuable for RATR. However, the initial phase of a complex HRRP, which is a function of the ratio of target distance to radar wavelength, is very sensitive to the variation of target's radial

* This work was partially supported by the National Science Foundation of China (NO.60402039) the National Defense Advanced Research Foundation of China (NO.51307060601).

D. Liu et al. (Eds.): ISNN 2007, Part I, LNCS 4491, pp. 1303–1309, 2007.

distance. Until now there has been little work in the field of the complex HRRP-based RATR. The literature [1] directly applied the adaptive Gaussian classifier (AGC), which can be used in the real HRRP-based RATR [3], to the complex HRRP-based RATR under the condition of using simulated turntable data. For simulated turntable data, the targets are assumed to only rotate around the turntable center in the simulation and the initial phase sensitivity problem need not be dealt with. However, in applications, at least the test samples are measured during the target's movement.

For high resolution radar system, we can get a complex echo vector consisting of the discrete echo components from different range cells. If the complex echo from one range cell is multiplied by the conjugate complex echo from another range cell, the initial phase will be counteracted and the corresponding difference phase, which is independent of the target position, will remain. By this property, a complex HRRP with some range cells' delay multiplied by the conjugate vector of the original complex HRRP will form a new complex feature vector with the difference phase information in the complex HRRP. And considering the influence of amplitude property on the recognition, each product term, i.e. each components in the above conjugate product vector, should be divided by a corresponding conversion factor. The recognition experiments based on measured data show that the proposed complex feature vector can obtain better recognition results than the real HRRP in the minimum Euclidean distance classifier and the AGC if only the cell interval parameters are proper.

2 Characteristic Comparison Between Real HRRP and Complex HRRP

As stated in the literatures [2,3,5], several issues have to be considered when real HRRP is applied to RATR, i.e. the target-aspect, time-shift and amplitude-scale sensitivities of real HRRP. Another important characteristic of real HRRP is that the average profile of a real HRRP frame can reduce the target-aspect sensitivity of the real HRRPs in a large extent [2].

Since the amplitude profile of a complex HRRP is the corresponding real HRRP, complex HRRP is also sensitive to the variation of target aspect, time shift and amplitude scale. For the target-aspect sensitivity, a set of complex HRRPs from a target-aspect sector without scatterers' motion through range cells (MTRC), is defined to be a complex HRRP frame, which can also be represented by a frame template. For the time-shift sensitivity, the time-shift compensation approaches in the training phase and in the test phase used in real HRRP-based RATR are also suitable for complex HRRP-based RATR. For the amplitude-scale sensitivity, complex HRRP should also be amplitude normalized before being used in RATR.

As discussed in Section 1, the initial phase of a complex HRRP is sensitive to the target distance variation. Similar to the time-shift sensitivity, there are also two kinds of initial phase sensitivity in RATR using complex HRRP. a) In the training phase, the initial phases of each training frame should be calibrated, which can be achieved by using the autofocus techniques in ISAR imaging. b) In the test phase, before a test sample is matched with a training frame, its initial phase should be calibrated with the training samples' initial phases. Obviously, similar to the time-shift compensation in the test phase, the autofocus techniques in ISAR imaging are not suitable for this process, since the test sample and the training frame may be from different targets. Actually, it is hard to deal with the initial phase sensitivity of complex HRRP in the test phase of complex HRRP-based RATR.

Because the average profile of a real HRRP frame can represent the stable scatterer auto-term (SAT) profile, the average profile of each frame can be used as the corresponding frame template in real HRRP-based RATR [2]. Then can the average profile of a complex HRRP frame be used as the corresponding frame template in complex HRRP-based RATR? Actually, after being amplitude normalized, the zero-Doppler slice in an ISAR image is identical with the average profile of a motion compensated complex HRRP frame, because the ISAR image is the Fourier transformation of the complex HRRP frame along the range cell dimension and its zero-Doppler channel output is theoretically identical with the average profile. The average profile of a real HRRP frame can represent the projection of the target's scatterer distribution onto the radial range dimension, while the average complex profile of a complex HRRP frame only represents the zero-Doppler slice of the target's scatterer distribution. Therefore, the average complex profile cannot be used as the corresponding frame template for complex HRRP-based RATR. Moreover, because the focus location in an ISAR image varies with the autofocus approach, the average complex profile of a complex HRRP frame is not exclusive. This further demonstrates that the average complex profile of a complex HRRP frame is not the suitable frame template for RATR using complex HRRPs.

Therefore, the recognition algorithms and frame-template-database establishment methods used in the real HRRP-based RATR can not be directly applied to the complex HRRP-based RATR.

3 A Novel Feature Extraction Method Using the Difference Phase Information in the Complex HRRP

3.1 The Fundamental Idea

Suppose there are radial displacement and little uniform rotation for a target. If the transmitted signal is $s(t)e^{j\omega_c t}$ with ω_c denoting the carrier angular frequency and the returned complex HRRP is denoted as a discrete complex vector, the m^{th} complex returned echo from the n^{th} range cell ($n=1,2,\cdots,N$) in baseband is

$$\begin{aligned} x_n(t,m) &= \sum_{i=1}^{L_n} \sigma_{ni} s\left(t-\frac{2R_{ni}(m)}{C}\right) e^{-j[\frac{4\pi}{\lambda}R_{ni}(m)]} \approx s\left(t-\frac{2R(m)}{c}\right)\sum_{i=1}^{L_n}\sigma_{ni} e^{-j\left\{\frac{4\pi}{\lambda}\left[R(m)+\Delta r_{ni}(m)\right]\right\}} \\ &= s\left(t-\frac{2R(m)}{c}\right) e^{j\psi(m)} \sum_{i=1}^{L_n}\sigma_{ni} e^{j\phi_{ni}(m)} \end{aligned} \tag{1}$$

where L_n is the number of target scatterers in the n^{th} range cell, σ_{ni} is the strength of the i^{th} scatterer in the n^{th} range cell, $R_{ni}(m)$ denotes the radial distance between the radar and the i^{th} scatterer in the n^{th} range cell in the m^{th} returned echo, $R(m)$ denotes the radial distance between the target reference center in the m^{th} returned echo and the radar, $\Delta r_{ni}(m)$ represents the radial displacement of the i th scatterer in the n th range cell in the m th returned echo, and $\psi(m) = -\frac{4\pi}{\lambda} R(m)$ denotes the initial phase of the m th returned echo. Without loss of generality, we assume that the transmitted signal $s(t)$ is a rectangular and narrow pulse with unit amplitude. After the time-shift compensation for the complex envelope, Eq. (1) can be written as

$$x_n(m) = e^{j\psi(m)} \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)} \tag{2}$$

Then the m th complex HRRP is

$$\begin{aligned} \tilde{\boldsymbol{x}}(m) &= \left[x_1(m), x_2(m), \cdots, x_n(m), \cdots, x_N(m)\right]^T \\ &= e^{j\psi(m)} \left[\sum_{i=1}^{L_1} \sigma_{1i} e^{j\phi_{1i}(m)}, \sum_{i=1}^{L_2} \sigma_{2i} e^{j\phi_{2i}(m)}, \cdots, \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)}, \cdots, \sum_{i=1}^{L_N} \sigma_{Ni} e^{j\phi_{Ni}(m)}\right]^T \end{aligned} \tag{3}$$

Obviously, the complex HRRP varies with the initial phase. According to Eqs. (2) and (3), for the complex echoes from all range cells in a complex HRRP, their initial phases are same. If the complex echo from the n_1 th range cell is multiplied by the conjugate complex echo from the n_2 th ($n_1 \neq n_2$) range cell, then

$$\begin{aligned} y_{n_1 n_2}(m) = x_{n_1}(m) \cdot x_{n_2}^*(m) &= \left[e^{j\psi(m)} \sum_{k=1}^{L_{n_1}} \sigma_{n_1 k} e^{j\phi_{n_1 k}(m)}\right] \cdot \left[e^{-j\psi(m)} \sum_{l=1}^{L_{n_2}} \sigma_{n_2 l} e^{j\phi_{n_2 l}(m)}\right] \\ &= \sum_{k=1}^{L_{n_1}} \sum_{l=1}^{L_{n_2}} \sigma_{n_1 k} \sigma_{n_2 l} \exp\left\{-j\frac{4\pi}{\lambda}\left[\Delta r_{n_1 k}(m) - \Delta r_{n_2 l}(m)\right]\right\} \end{aligned} \tag{4}$$

Obviously, the product term $y_{n_1 n_2}(m)$ is independent of the initial phase. Let $L_{n_1 n_2} = L_{n_1} \cdot L_{n_2}$, $\sigma_{n_1 n_2 i} = \sigma_{n_1 k} \cdot \sigma_{n_2 l}$, $\Delta r_{n_1 n_2 i}(m) = \Delta r_{n_1 k}(m) - \Delta r_{n_2 l}(m)$. Then

$$y_{n_1 n_2}(m) = \sum_{i=1}^{L_{n_1 n_2}} \sigma_{n_1 n_2 i} \exp\left[-j\frac{4\pi}{\lambda} \Delta r_{n_1 n_2 i}(m)\right] = \sum_{i=1}^{L_{n_1 n_2}} \sigma_{n_1 n_2 i} e^{j\phi_{n_1 n_2 i}(m)} \tag{5}$$

The product term $y_{n_1 n_2}(m)$ can also be regarded as the complex echo from a suppositional range cell in a special kind of complex HRRP, which is the vector product of the complex HRRP with J ($n_2 - n_1 = J$) range cells' delay and the conjugate vector of the original complex HRRP. If x_{ni} and y_{ni} represent the initial cross and radial coordinates of the i th scatterer in the n th range cell, then

$$\begin{aligned} y_{n_1 n_2}(m) &= \sum_{k=1}^{L_{n_1}} \sum_{l=1}^{L_{n_2}} \sigma_{n_1 k} \sigma_{n_2 l} \exp\left\{-j\frac{4\pi}{\lambda}\left\{\begin{aligned}&\left(y_{n_1 k} - y_{n_2 l}\right) \cdot \left[\cos(m \cdot \Delta\phi) - 1\right] \\ &+ \left(x_{n_1 k} - x_{n_2 l}\right) \sin(m \cdot \Delta\phi)\end{aligned}\right\}\right\} \\ &\overset{\substack{y_{n_1 k} \approx y_{n_1} \\ y_{n_2 k} \approx y_{n_2}}}{\approx} \sum_{k=1}^{L_{n_1}} \sum_{l=1}^{L_{n_2}} \sigma_{n_1 k} \sigma_{n_2 l} \exp\left\{-j\frac{4\pi}{\lambda}\left\{\begin{aligned}&\left(y_{n_1} - y_{n_2}\right) \cdot \left[\cos(m \cdot \Delta\phi) - 1\right] \\ &+ \left(x_{n_1 k} - x_{n_2 l}\right) \sin(m \cdot \Delta\phi)\end{aligned}\right\}\right\} \\ &= \sum_{k=1}^{L_{n_1}} \sum_{l=1}^{L_{n_2}} \sigma_{n_1 k} \sigma_{n_2 l} \left\{\begin{aligned}&\cos\left[\theta_{n_1 n_2}(m) + \varphi_{n_1 k n_2 l}(m)\right] \\ &+ j\sin\left[\theta_{n_1 n_2}(m) + \varphi_{n_1 k n_2 l}(m)\right]\end{aligned}\right\} \end{aligned} \tag{6}$$

where $\Delta\phi$ denotes the target-aspect difference between two neighboring echoes. When J is small, $\theta_{n_1 n_2}(m) \ll \varphi_{n_1 k n_2 l}(m)$ and $\theta_{n_1 n_2}(m)$ can be ignored. Then

$$y_{n_1 n_2}(m) \approx \sum_{k=1}^{L_{n_1}} \sum_{l=1}^{L_{n_2}} \sigma_{n_1 k} \sigma_{n_2 l} \exp\left[j\varphi_{n_1 k n_2 l}(m) \right] = \sum_{k=1}^{L_{n_1}} \sum_{l=1}^{L_{n_2}} \sigma_{n_1 k} \sigma_{n_2 l} \left\{ \cos\left[\varphi_{n_1 k n_2 l}(m) \right] + j \sin\left[\varphi_{n_1 k n_2 l}(m) \right] \right\} \quad (7)$$

The product term $y_{n_1 n_2}(m)$ in Eq. (7) depends on the cross coordinates' differences between the scatterers in the n_1 th and the n_2 th range cells. Then the physical mechanisms of the product term are similar to those of the power term of the real HRRP [2]. The above feature extraction method uses the conjugate product between the complex echoes from two range cells with J intervals (J is small) to counteract the initial phase, yet the feature vector is in power form, and because of the amplitudes of the two complex echoes being similar, the amplitude profile of the product vector can increase the amplitude's dynamic range of the original complex HRRP, reduce the contribution of weak scatterers to the similarity measurement, and enlarge the speckle effect. According to the literatures [3~5], this is unfavorable for the recognition from the viewpoint of using the HRRP's amplitude property. In order to convert the conjugate product vector into the one in amplitude form, we propose a novel feature vector

$$\begin{aligned} z_J(m) &= \left[z_{J1}(m), z_{J2}(m), \cdots, z_{Jn}(m), \cdots, z_{J(N-J)}(m) \right]^T \\ &= \left[\frac{x_1(m) \cdot x_{1+J}^*(m)}{\sqrt{\left|x_1(m)\right|_2^2 + \left|x_{1+J}^*(m)\right|_2^2}}, \frac{x_2(m) \cdot x_{2+J}^*(m)}{\sqrt{\left|x_2(m)\right|_2^2 + \left|x_{2+J}^*(m)\right|_2^2}}, \cdots, \right. \\ &\qquad \left. \frac{x_n(m) \cdot x_{n+J}^*(m)}{\sqrt{\left|x_n(m)\right|_2^2 + \left|x_{n+J}^*(m)\right|_2^2}}, \cdots, \frac{x_{N-J}(m) \cdot x_N^*(m)}{\sqrt{\left|x_{N-J}(m)\right|_2^2 + \left|x_N^*(m)\right|_2^2}} \right]^T \end{aligned} \quad (8)$$

For both large echoes, their product term is divided by a large conversion factor, while for both small echoes, their product term is divided by a small conversion factor, therefore, such feature extraction method can reduce the amplitude's dynamic range of the product vector and the amplitude property of the complex feature vector is more favorable for recognition. The complex feature vector in Eq. (8) is referred to as the complex feature vector with difference phases in this paper. According to what discussed above, such complex feature vector can be regarded as a complex echo vector consisting of the discrete echo components from the suppositional range cells. The amplitude property of such feature vector is more favorable for recognition. If the complex feature vectors extracted from all training and test samples by our feature extraction method are applied to RATR, due to containing no initial phases of the complex HRRPs, the recognition algorithms, frame-template-database establishment methods and preprocessing methods used in the real HRRP-based RATR can also be applied to the proposed complex feature vector-based RATR.

3.2 The Recognition Algorithm

According to the idea in Section 3.1, the complex feature vector with difference phases is the product vector divided by the corresponding conversion factors, therefore,

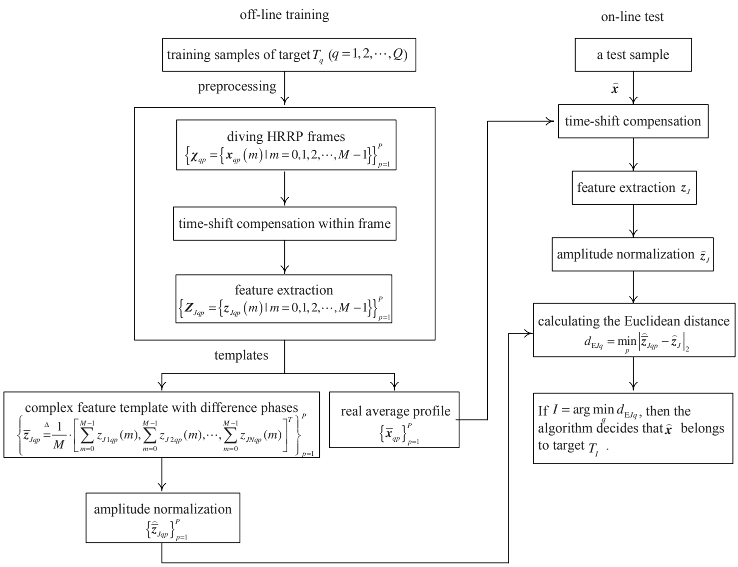


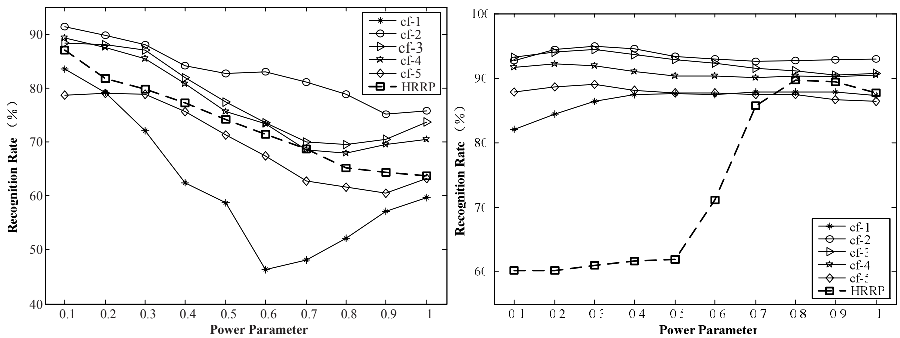
Fig. 1. The flow chart of the minimum Euclidean distance classifier using the proposed featureextraction method

the range cell order of a complex HRRP will influence the feature extraction. However, due to the time-shift sensitivity of an HRRP, which has been discussed in Section 2, the range cell order of an HRRP varies with the range window, and all of the samples used in the training and test phases should be time-shift compensated. To solve the time-shift sensitivity, all of the complex HRRPs used in the training and test phases are time-shift compensated by the slide correlation processing in our recognition algorithm, thus the complex feature vectors with difference phases extracted from the time-shift compensated complex HRRPs are independent of the time shift and can be used to classifiers directly. Fig. 1 takes the minimum Euclidean distance classifier as an example to show how to apply the feature extraction method proposed in this paper to classifiers.

4 Experimental Results

The results presented in this paper are based on real airplane data, which are measured by radar with the center frequency of $5520MHz$ and the bandwidth of $400MHz$. Training data cover almost all of the target-aspect angles of test data, but their elevation angles are different.

Fig. 2 (a) and (b) shows the recognition rates of the proposed complex feature vectors with $J=1,2,3,4,5$ and the real HRRP versus the power parameter α in the minimum Euclidean distance classifier [2,4] and the AGC [1,3]. The recognition experiments based on measured data show that the proposed feature vector can obtain better recognition results than the real HRRP in the minimum Euclidean distance classifier and the AGC if only the cell interval parameters are proper.



(a) The minimum Euclidean distance classifier (b) The AGC

Fig. 2. The average recognition rates of the real HRRP and the complex feature vectors with $J = 1,2,3,4,5$ versus the power parameter in the minimum Euclidean distance classifier and the AGC

5 Conclusion

A novel feature extraction method which can counteract the initial phase and leave the difference phase information in the complex HRRP is proposed in this paper. The recognition experiments based on measured data show that the proposed feature extraction method can effectively use the difference phase information in the complex HRRP and improve the recognition performance.

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